Ex - 4

# SETTING UP A LOCAL AREA NETWORK

AIM:

To design set up, and configure a Local Area Network (LAN) connecting four computers through a network switch. assign static IP address to all devices, verify connectivity using the ping command, and demonstrate resource / data sharing across the network

~~EQW~~

PROCEDURE:

Since physical hardware was limited, the LAN was also implemented and verified using Cisco Packet Tracer simulation software, following the steps below

STEP 1: Place the Devices

1. Open Cisco Packet Tracer
2. From the device category panel, select END Devices and drag 4 PCs onto the workspace.
3. Select Switches and drag 1 switch onto the workspace

STEP 2: Cable the Devices

4. Click Connections → select Copper Straight-Through cable
5. Click each PC, then click the switch and select an available FastEthernet port to complete each connection.
6. Repeat for all four PCs. Green link lights confirm successful physical connection

STEP 3: Assign IP Addresses to PCs

7. Click a PC → Desktop tab → IP Configuration
8. Enter the Static IP Address and Subnet Mask as per the IP table

STEP 4:

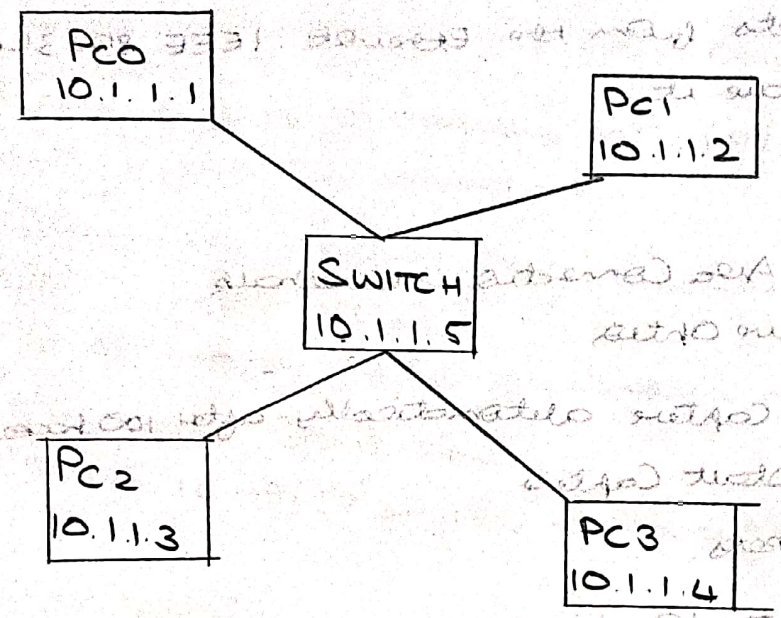
9. Click the switch → CLI tab
10. enable
    configure terminal
    interface vlan 1
    ip address 10.1.1.5 255.0.0.0

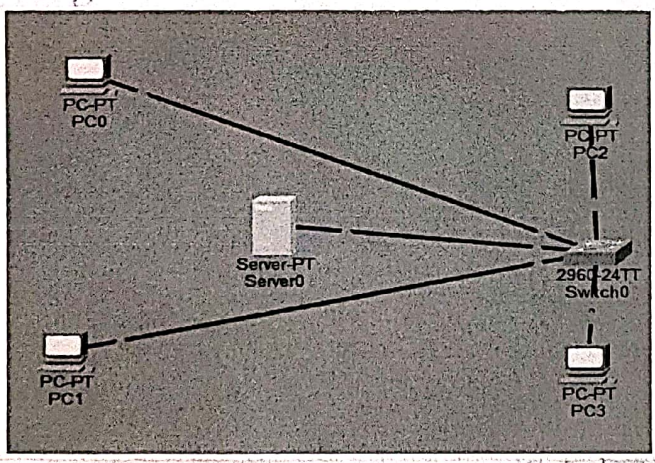
STEP 5 Verify Connectivity Using Ping

STEP 6 Demostrate Data Sharing

STEP 7 Observe Packet flow in Simulation Mode

STAR TOPOLOGY LAN




RESULT

Successfully connected all four PCs to the switch in a star topology and tested ping between all devices.